

Source folder: /Users/jonat/Documents/GitHub/master_thesis
Files in this part: 17

File: results_backup/dp_inf_alpha_1.0_dim_64_clients_100_seed_456.json

```
{
  "experiment_id": "dp_inf_alpha_1.0_dim_64_clients_100_seed_456",
  "timestamp": "2026-02-11T03:05:31.166801",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 1.0,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 13.72330759866843,
      "test_metrics": {
        "mse": 13.688510116577149,
        "mae": 3.5223534229278566,
        "samples": 20000,
        "NDCG@10": 0.0577968700397596,
        "Hit@10": 0.09,
        "Precision@10": 0.009,
        "Recall@10": 0.0037602713721248985,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 13.719931132283033,
      "test_metrics": {
        "mse": 13.68849630279541,
        "mae": 3.522351094818115,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 3,
      "train_loss": 13.726395302521766,
      "test_metrics": {
        "mse": 13.688480722045899,
        "mae": 3.5223484970092773,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 4,
      "train_loss": 13.716910944022791,
      "test_metrics": {
        "mse": 13.688465667724609,
```

```
"mae": 3.5223459297180177,
"samples": 20000
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 5,
  "train_loss": 13.724981207493943,
  "test_metrics": {
    "mse": 13.688449787902831,
    "mae": 3.522343238067627,
    "samples": 20000,
    "NDCG@10": 0.0577968700397596,
    "Hit@10": 0.09,
    "Precision@10": 0.009,
    "Recall@10": 0.0037602713721248985,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 13.716727754199491,
  "test_metrics": {
    "mse": 13.688433050537109,
    "mae": 3.522340389251709,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 13.717107462780824,
  "test_metrics": {
    "mse": 13.688415927124023,
    "mae": 3.5223376194000244,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.721387370022974,
  "test_metrics": {
    "mse": 13.68839663848877,
    "mae": 3.522334504699707,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 13.718756730316883,
  "test_metrics": {
    "mse": 13.688377203369141,
    "mae": 3.5223312973022463,
    "samples": 20000
  },
  "aggregation": {
```

```

        "num_clients": 100,
        "total_samples": 80000
    },
    {
        "round": 10,
        "train_loss": 13.718138670994527,
        "test_metrics": {
            "mse": 13.68835595703125,
            "mae": 3.5223278732299805,
            "samples": 20000,
            "NDCG@10": 0.057916521733220525,
            "Hit@10": 0.09,
            "Precision@10": 0.009,
            "Recall@10": 0.0037602713721248985,
            "num_users_evaluated": 100
        },
        "aggregation": {
            "num_clients": 100,
            "total_samples": 80000
        }
    }
],
"final_metrics": {
    "mse": 13.68835595703125,
    "mae": 3.5223278732299805,
    "samples": 20000,
    "NDCG@10": 0.057916521733220525,
    "Hit@10": 0.09,
    "Precision@10": 0.009,
    "Recall@10": 0.0037602713721248985,
    "num_users_evaluated": 100
}
}

```

File: results_backup/dp_inf_alpha_1.0_dim_64_clients_100_seed_456_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.72330759866843,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688510116577149,3.52235342292785
66
2,13.719931132283033,,,,,13.68849630279541,3.522351094818115
3,13.726395302521766,,,,,13.688480722045899,3.5223484970092773
4,13.716910944022791,,,,,13.688465667724609,3.5223459297180177
5,13.724981207493943,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688449787902831,3.5223432380676
27
6,13.716727754199491,,,,,13.688433050537109,3.522340389251709
7,13.717107462780824,,,,,13.688415927124023,3.5223376194000244
8,13.721387370022974,,,,,13.68839663848877,3.522334504699707
9,13.718756730316883,,,,,13.688377203369141,3.5223312973022463
10,13.718138670994527,0.057916521733220525,0.09,0.009,0.0037602713721248985,13.68835595703125,3.52232787322
99805
FINAL,,0.057916521733220525,0.09,0.009,0.0037602713721248985,13.68835595703125,3.5223278732299805

```

File: results_old_broken/centralized_baseline.json

```

{
    "experiment_id": "centralized_baseline",
    "config": {
        "num_users": 50,
        "num_items": 4032,
        "embedding_dim": 16,
        "num_epochs": 50,
        "learning_rate": 0.01,
        "batch_size": 64,
        "seed": 42,
        "train_samples": 6901,
        "test_samples": 1726
    },
    "train_losses": [
        0.4571514339358718,
        0.4572106903349912,
        0.45723974815121404,
        0.45714877683807303,
        0.4571779350991602,

```

```
0.4573572344250149,
0.4571161896542267,
0.4572953239635185,
0.4571742277454447,
0.45738347702556187,
0.4572924866720482,
0.45717150690378966,
0.4572609251847974,
0.4572597457854836,
0.4571085817835949,
0.45722793263417705,
0.45692682376614324,
0.45725621393433324,
0.45722514749677096,
0.4571343767974112,
0.457103347060857,
0.4572525965394797,
0.4571914984672158,
0.45731091609707586,
0.45736978810142587,
0.45721873116713985,
0.4571277319833084,
0.4569769333358164,
0.4571558488187967,
0.4571550753381517,
0.4570941574595593,
0.4571533015480748,
0.4572122320532799,
0.4571514888494103,
0.45724067009157604,
0.45711952199538547,
0.457238949559353,
0.4571778189252924,
0.4572068919186239,
0.4572361194425159,
0.45708482270991363,
0.4570839548552478,
0.4572329703304503,
0.45729242899903544,
0.4574113871764254,
0.4571401521011635,
0.45713934633466935,
0.4571683508930383,
0.45707717731043146,
0.45704644200978456
],
"test_metrics_history": [
  {
    "epoch": 10,
    "metrics": {
      "mse": 0.4507715737971619,
      "mae": 0.45093746671532975,
      "accuracy": 0.5492468134414832,
      "samples": 1726,
      "NDCG@10": 0.0,
      "Hit@10": 0.04,
      "Precision@10": 0.004,
      "Recall@10": 0.000744215134459037,
      "num_users_evaluated": 50
    }
  },
  {
    "epoch": 20,
    "metrics": {
      "mse": 0.45077231723215183,
      "mae": 0.45093778387005606,
      "accuracy": 0.5492468134414832,
      "samples": 1726,
      "NDCG@10": 0.0,
      "Hit@10": 0.04,
      "Precision@10": 0.004,
      "Recall@10": 0.000744215134459037,
      "num_users_evaluated": 50
    }
  }
]
```

```

    },
    {
      "epoch": 30,
      "metrics": {
        "mse": 0.45077307358263263,
        "mae": 0.45093865797942373,
        "accuracy": 0.5492468134414832,
        "samples": 1726,
        "NDCG@10": 0.0,
        "Hit@10": 0.04,
        "Precision@10": 0.004,
        "Recall@10": 0.000744215134459037,
        "num_users_evaluated": 50
      }
    },
    {
      "epoch": 40,
      "metrics": {
        "mse": 0.4507738197112277,
        "mae": 0.45094001942410256,
        "accuracy": 0.5492468134414832,
        "samples": 1726,
        "NDCG@10": 0.0,
        "Hit@10": 0.04,
        "Precision@10": 0.004,
        "Recall@10": 0.000744215134459037,
        "num_users_evaluated": 50
      }
    },
    {
      "epoch": 50,
      "metrics": {
        "mse": 0.4507745841425236,
        "mae": 0.45094192677273537,
        "accuracy": 0.5492468134414832,
        "samples": 1726,
        "NDCG@10": 0.0,
        "Hit@10": 0.04,
        "Precision@10": 0.004,
        "Recall@10": 0.000744215134459037,
        "num_users_evaluated": 50
      }
    }
  ],
  "final_metrics": {
    "mse": 0.4507745841425236,
    "mae": 0.45094192677273537,
    "accuracy": 0.5492468134414832,
    "samples": 1726,
    "NDCG@10": 0.0,
    "Hit@10": 0.04,
    "Precision@10": 0.004,
    "Recall@10": 0.000744215134459037,
    "num_users_evaluated": 50
  }
}

```

File: results_old_broken/dp_1_alpha_0.5_dim_16_clients_100_seed_123.json

```

{
  "experiment_id": "dp_1_alpha_0.5_dim_16_clients_100_seed_123",
  "timestamp": "2025-12-02T02:29:40.281723",
  "config": {
    "num_users": 50,
    "num_items": 4032,
    "embedding_dim": 16,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 1,
    "use_dp": true,
    "dp_sigma": 1.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 1,

```

```
"learning_rate": 0.01,
"batch_size": 32,
"seed": 123
},
"rounds": [
{
  "round": 1,
  "train_loss": 0.46367318129430807,
  "test_metrics": {
    "mse": 0.4507430170944364,
    "mae": 0.4509384922191138,
    "accuracy": 0.5492468134414832,
    "samples": 1726,
    "NDCG@10": 0.0,
    "Hit@10": 0.08,
    "Precision@10": 0.008,
    "Recall@10": 0.002240313197239193,
    "num_users_evaluated": 50
  },
  "aggregation": {
    "status": "aggregated",
    "round": 130,
    "num_clients": 97,
    "total_samples": 6706
  },
  "client_metrics": [
    {
      "client_id": "client_1",
      "loss": 0.5557109021162439,
      "samples": 60,
      "epochs": 1
    },
    {
      "client_id": "client_2",
      "loss": 0.6457548787007009,
      "samples": 53,
      "epochs": 1
    },
    {
      "client_id": "client_3",
      "loss": 0.6056538390877787,
      "samples": 92,
      "epochs": 1
    },
    {
      "client_id": "client_4",
      "loss": 0.4426552824222062,
      "samples": 56,
      "epochs": 1
    },
    {
      "client_id": "client_5",
      "loss": 0.4980737694841382,
      "samples": 81,
      "epochs": 1
    },
    {
      "client_id": "client_6",
      "loss": 0.4532010077401638,
      "samples": 64,
      "epochs": 1
    },
    {
      "client_id": "client_7",
      "loss": 0.32106738017764114,
      "samples": 75,
      "epochs": 1
    },
    {
      "client_id": "client_8",
      "loss": 0.3913428028577159,
      "samples": 122,
      "epochs": 1
    }
  ],
}
```

```
{
  "client_id": "client_9",
  "loss": 0.3203915064833753,
  "samples": 98,
  "epochs": 1
},
{
  "client_id": "client_10",
  "loss": 0.5723679045291749,
  "samples": 66,
  "epochs": 1
},
{
  "client_id": "client_11",
  "loss": 0.40603614413273054,
  "samples": 40,
  "epochs": 1
},
{
  "client_id": "client_12",
  "loss": 0.4035525654183543,
  "samples": 105,
  "epochs": 1
},
{
  "client_id": "client_13",
  "loss": 0.6469241126303203,
  "samples": 42,
  "epochs": 1
},
{
  "client_id": "client_14",
  "loss": 0.38748478887645404,
  "samples": 86,
  "epochs": 1
},
{
  "client_id": "client_15",
  "loss": 0.6464829872809903,
  "samples": 68,
  "epochs": 1
},
{
  "client_id": "client_16",
  "loss": 0.41881146399390223,
  "samples": 136,
  "epochs": 1
},
{
  "client_id": "client_18",
  "loss": 0.18751411161477058,
  "samples": 34,
  "epochs": 1
},
{
  "client_id": "client_19",
  "loss": 0.3818149188003666,
  "samples": 82,
  "epochs": 1
},
{
  "client_id": "client_20",
  "loss": 0.5144551867022953,
  "samples": 86,
  "epochs": 1
},
{
  "client_id": "client_21",
  "loss": 0.5387382644853522,
  "samples": 63,
  "epochs": 1
},
{
  "client_id": "client_22",
```

```
"loss": 0.459884454077768,
"samples": 101,
"epochs": 1
},
{
  "client_id": "client_23",
  "loss": 0.37488207639301674,
  "samples": 70,
  "epochs": 1
},
{
  "client_id": "client_24",
  "loss": 0.4380109538911902,
  "samples": 91,
  "epochs": 1
},
{
  "client_id": "client_25",
  "loss": 0.4548496875939477,
  "samples": 73,
  "epochs": 1
},
{
  "client_id": "client_26",
  "loss": 0.2916615439482533,
  "samples": 70,
  "epochs": 1
},
{
  "client_id": "client_27",
  "loss": 0.34939993198795855,
  "samples": 81,
  "epochs": 1
},
{
  "client_id": "client_28",
  "loss": 0.5814488901170008,
  "samples": 57,
  "epochs": 1
},
{
  "client_id": "client_29",
  "loss": 0.4843779764618296,
  "samples": 40,
  "epochs": 1
},
{
  "client_id": "client_30",
  "loss": 0.4530479579900667,
  "samples": 44,
  "epochs": 1
},
{
  "client_id": "client_31",
  "loss": 0.583445513475112,
  "samples": 68,
  "epochs": 1
},
{
  "client_id": "client_32",
  "loss": 0.32801371803411844,
  "samples": 48,
  "epochs": 1
},
{
  "client_id": "client_33",
  "loss": 0.44745833660949935,
  "samples": 66,
  "epochs": 1
},
{
  "client_id": "client_34",
  "loss": 0.5937374692779251,
  "samples": 36,
```



```
"epochs": 1
},
{
  "client_id": "client_35",
  "loss": 0.5655865523526741,
  "samples": 42,
  "epochs": 1
},
{
  "client_id": "client_36",
  "loss": 0.30546544244170765,
  "samples": 70,
  "epochs": 1
},
{
  "client_id": "client_37",
  "loss": 0.5142286232911143,
  "samples": 55,
  "epochs": 1
},
{
  "client_id": "client_38",
  "loss": 0.48646679199619197,
  "samples": 78,
  "epochs": 1
},
{
  "client_id": "client_39",
  "loss": 0.49770812491264504,
  "samples": 84,
  "epochs": 1
},
{
  "client_id": "client_40",
  "loss": 0.37963131265108985,
  "samples": 99,
  "epochs": 1
},
{
  "client_id": "client_41",
  "loss": 0.428615933835862,
  "samples": 71,
  "epochs": 1
},
{
  "client_id": "client_42",
  "loss": 0.5001916211481319,
  "samples": 66,
  "epochs": 1
},
{
  "client_id": "client_43",
  "loss": 0.450505144026546,
  "samples": 109,
  "epochs": 1
},
{
  "client_id": "client_44",
  "loss": 0.5107724156113636,
  "samples": 68,
  "epochs": 1
},
{
  "client_id": "client_45",
  "loss": 0.2976898253797516,
  "samples": 119,
  "epochs": 1
},
{
  "client_id": "client_46",
  "loss": 0.4890568493302942,
  "samples": 46,
  "epochs": 1
},
},
```

```
{
  "client_id": "client_47",
  "loss": 0.5313180263627382,
  "samples": 50,
  "epochs": 1
},
{
  "client_id": "client_48",
  "loss": 0.6669454338255008,
  "samples": 68,
  "epochs": 1
},
{
  "client_id": "client_49",
  "loss": 0.27081364880331016,
  "samples": 68,
  "epochs": 1
},
{
  "client_id": "client_50",
  "loss": 0.5382756572504659,
  "samples": 26,
  "epochs": 1
},
{
  "client_id": "client_51",
  "loss": 0.4841822069402118,
  "samples": 106,
  "epochs": 1
},
{
  "client_id": "client_52",
  "loss": 0.6719577975994017,
  "samples": 40,
  "epochs": 1
},
{
  "client_id": "client_53",
  "loss": 0.42807876792594857,
  "samples": 111,
  "epochs": 1
},
{
  "client_id": "client_54",
  "loss": 0.42539091000332885,
  "samples": 58,
  "epochs": 1
},
{
  "client_id": "client_55",
  "loss": 0.39591079667645784,
  "samples": 80,
  "epochs": 1
},
{
  "client_id": "client_56",
  "loss": 0.3980079091350835,
  "samples": 51,
  "epochs": 1
},
{
  "client_id": "client_57",
  "loss": 0.4688365088746671,
  "samples": 84,
  "epochs": 1
},
{
  "client_id": "client_58",
  "loss": 0.6204763406595236,
  "samples": 29,
  "epochs": 1
},
{
  "client_id": "client_59",
```

```
"loss": 0.4448296438752705,
"samples": 54,
"epochs": 1
},
{
  "client_id": "client_60",
  "loss": 0.2577935043561469,
  "samples": 98,
  "epochs": 1
},
{
  "client_id": "client_61",
  "loss": 0.33273017161232354,
  "samples": 45,
  "epochs": 1
},
{
  "client_id": "client_62",
  "loss": 0.5360660355548924,
  "samples": 75,
  "epochs": 1
},
{
  "client_id": "client_63",
  "loss": 0.4122363117906802,
  "samples": 109,
  "epochs": 1
},
{
  "client_id": "client_64",
  "loss": 0.34019174220690784,
  "samples": 172,
  "epochs": 1
},
{
  "client_id": "client_65",
  "loss": 0.5436274373365956,
  "samples": 53,
  "epochs": 1
},
{
  "client_id": "client_66",
  "loss": 0.5521192597742065,
  "samples": 80,
  "epochs": 1
},
{
  "client_id": "client_67",
  "loss": 0.44971266255001785,
  "samples": 42,
  "epochs": 1
},
{
  "client_id": "client_68",
  "loss": 0.45814414448771335,
  "samples": 80,
  "epochs": 1
},
{
  "client_id": "client_69",
  "loss": 0.43108936451436924,
  "samples": 42,
  "epochs": 1
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  "samples": 70,
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{
  "client_id": "client_80",
  "loss": 0.5102154103916059,
  "samples": 70,
  "epochs": 1
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{
  "client_id": "client_81",
  "loss": 0.4219775772211265,
  "samples": 56,
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{
  "client_id": "client_82",
  "loss": 0.48447641348345766,
  "samples": 48,
  "epochs": 1
},
{
  "client_id": "client_83",
  "loss": 0.5336338887037632,
  "samples": 69,
  "epochs": 1
},
{
  "client_id": "client_84",
  "loss": 0.3543699493936462,
  "samples": 96,
  "epochs": 1
},
{
  "client_id": "client_85",
  "loss": 0.5830394988721725,
  "samples": 41,
  "epochs": 1
},
{
  "client_id": "client_86",
  "loss": 0.5281228762736448,
  "samples": 52,
  "epochs": 1
},
{
  "client_id": "client_87",
  "loss": 0.598926632156328,
  "samples": 35,
  "epochs": 1
},
{
  "client_id": "client_88",
  "loss": 0.5081159465083185,
  "samples": 51,
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    "epochs": 1
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  {
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    "loss": 0.46157241376439556,
    "samples": 71,
    "epochs": 1
  },
  {
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    "loss": 0.2829368363441543,
    "samples": 85,
    "epochs": 1
  },
  {
    "client_id": "client_91",
    "loss": 0.5445289783541304,
    "samples": 67,
    "epochs": 1
  },
  {
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    "loss": 0.37506953499845846,
    "samples": 64,
    "epochs": 1
  },
  {
    "client_id": "client_93",
    "loss": 0.6384704496933029,
    "samples": 55,
    "epochs": 1
  },
  {
    "client_id": "client_94",
    "loss": 0.5502579143108164,
    "samples": 37,
    "epochs": 1
  },
  {
    "client_id": "client_95",
    "loss": 0.7169195703293536,
    "samples": 46,
    "epochs": 1
  },
  {
    "client_id": "client_96",
    "loss": 0.5241496603931572,
    "samples": 70,
    "epochs": 1
  },
  {
    "client_id": "client_97",
    "loss": 0.502627613166212,
    "samples": 49,
    "epochs": 1
  },
  {
    "client_id": "client_98",
    "loss": 0.529231908340493,
    "samples": 85,
    "epochs": 1
  },
  {
    "client_id": "client_99",
    "loss": 0.3403145803107575,
    "samples": 76,
    "epochs": 1
  }
],
"timestamp": "2025-12-02T04:13:16.517318"
}
],
"final_metrics": {
  "mse": 0.4507438939664759,
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    "mae": 0.45100273649496386,  
    "accuracy": 0.5492468134414832,  
    "samples": 1726,  
    "NDCG@10": 0.0,  
    "Hit@10": 0.04,  
    "Precision@10": 0.004,  
    "Recall@10": 0.0007066997518610422,  
    "num_users_evaluated": 50  
  },  
  "client_metrics": [],  
  "mobile_metrics": []  
}
```

File: results_old_broken/dp_1_alpha_0.5_dim_16_clients_100_seed_123_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients>Total_Samples  
1,0.46367318129430807,0.0,0.08,0.008,0.002240313197239193,0.4507430170944364,0.4509384922191138,0.549246813  
4414832,97,6706  
2,0.4684241585107927,0.0,0.06,0.006000000000000001,0.001595151906916613,0.4507414323568068,0.45094553150694  
17,0.5492468134414832,97,6706  
3,0.4706975849528031,0.0,0.04,0.008,0.0013610859728506789,0.45074376481156775,0.45095127896942105,0.5492468  
134414832,97,6706  
4,0.4719573202744622,0.0,0.06,0.008,0.0014608378338928623,0.4507482812274787,0.45095780108508204,0.54924681  
34414832,97,6706  
5,0.4763704327345546,0.0,0.06,0.01,0.0020062472631732593,0.4507509999037618,0.45096745054525683,0.549246813  
4414832,97,6706  
6,0.4673733066272844,0.0,0.08,0.008,0.0022279979263032327,0.4507485800104373,0.45097457381830847,0.54924681  
34414832,97,6706  
7,0.4739746963003365,0.0,0.1,0.01,0.0027080861650121612,0.45074725434235874,0.45097843934873577,0.549246813  
4414832,97,6706  
8,0.47487636073820433,0.0,0.14,0.014000000000000002,0.0034876963599147098,0.4507421903648907,0.450991655970  
78006,0.5492468134414832,97,6706  
9,0.4787405678631876,0.0,0.12,0.012000000000000002,0.0030801907110297507,0.45074208103215296,0.450995883963  
7518,0.5492468134414832,97,6706  
10,0.46736729071438465,0.0,0.04,0.004,0.0007066997518610422,0.4507438939664759,0.45100273649496386,0.549246  
8134414832,97,6706  
FINAL,,0.0,0.04,0.004,0.0007066997518610422,0.4507438939664759,0.45100273649496386,0.5492468134414832,,
```

File: results_old_broken/dp_1_alpha_0.5_dim_16_clients_100_seed_42.json

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{  
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  "timestamp": "2025-12-02T00:46:22.288464",  
  "config": {  
    "num_users": 50,  
    "num_items": 4032,  
    "embedding_dim": 16,  
    "num_clients": 100,  
    "alpha": 0.5,  
    "dp_epsilon": 1,  
    "use_dp": true,  
    "dp_sigma": 1.0,  
    "dp_clip_norm": 1.0,  
    "num_rounds": 10,  
    "local_epochs": 1,  
    "learning_rate": 0.01,  
    "batch_size": 32,  
    "seed": 42  
  },  
  "rounds": [  
    {  
      "round": 1,  
      "train_loss": 0.4653022545728739,  
      "test_metrics": {  
        "mse": 0.45074916921927227,  
        "mae": 0.45093584889470495,  
        "accuracy": 0.5492468134414832,  
        "samples": 1726,  
        "NDCG@10": 0.0,  
        "Hit@10": 0.1,  
        "Precision@10": 0.01,  
        "Recall@10": 0.008234543738859958,  
        "num_users_evaluated": 50  
      }  
    }  
  ],  
}
```



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"aggregation": {
  "status": "aggregated",
  "round": 120,
  "num_clients": 97,
  "total_samples": 6706
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"client_metrics": [
  {
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    "loss": 0.5490942236103894,
    "samples": 60,
    "epochs": 1
  },
  {
    "client_id": "client_2",
    "loss": 0.6542810208495256,
    "samples": 53,
    "epochs": 1
  },
  {
    "client_id": "client_3",
    "loss": 0.6101096225591317,
    "samples": 92,
    "epochs": 1
  },
  {
    "client_id": "client_4",
    "loss": 0.4477461572905184,
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    "loss": 0.4980809816437519,
    "samples": 81,
    "epochs": 1
  },
  {
    "client_id": "client_6",
    "loss": 0.4531704454738364,
    "samples": 64,
    "epochs": 1
  },
  {
    "client_id": "client_7",
    "loss": 0.340811129056177,
    "samples": 75,
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  {
    "client_id": "client_8",
    "loss": 0.3912935172793886,
    "samples": 122,
    "epochs": 1
  },
  {
    "client_id": "client_9",
    "loss": 0.32042706273933413,
    "samples": 98,
    "epochs": 1
  },
  {
    "client_id": "client_10",
    "loss": 0.4165828487564344,
    "samples": 66,
    "epochs": 1
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  {
    "client_id": "client_11",
    "loss": 0.5003760488356038,
    "samples": 40,
    "epochs": 1
  },
  {

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"client_id": "client_12",
"loss": 0.4235767259234455,
"samples": 105,
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{
  "client_id": "client_13",
  "loss": 0.5783833949666277,
  "samples": 42,
  "epochs": 1
},
{
  "client_id": "client_14",
  "loss": 0.4016459536850269,
  "samples": 86,
  "epochs": 1
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{
  "client_id": "client_15",
  "loss": 0.7190672543854187,
  "samples": 68,
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  "client_id": "client_16",
  "loss": 0.40003334627812076,
  "samples": 136,
  "epochs": 1
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  "loss": 0.18758367830759062,
  "samples": 34,
  "epochs": 1
},
{
  "client_id": "client_19",
  "loss": 0.3979743087907445,
  "samples": 82,
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{
  "client_id": "client_20",
  "loss": 0.4859268237650537,
  "samples": 86,
  "epochs": 1
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  "client_id": "client_21",
  "loss": 0.5404358337220931,
  "samples": 63,
  "epochs": 1
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{
  "client_id": "client_22",
  "loss": 0.4173788986734161,
  "samples": 101,
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  "samples": 70,
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  "client_id": "client_24",
  "loss": 0.4377466073797268,
  "samples": 91,
  "epochs": 1
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{
  "client_id": "client_25",
  "loss": 0.5346942847831744,
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"samples": 73,
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  "loss": 0.38212364095507334,
  "samples": 70,
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  "loss": 0.3770619069328098,
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  "samples": 57,
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  "samples": 68,
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  "samples": 48,
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  "loss": 0.44731849422536296,
  "samples": 66,
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{
  "client_id": "client_34",
  "loss": 0.702959096619588,
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  "loss": 0.3506559852156797,
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{
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  "samples": 55,
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  "samples": 84,
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  "loss": 0.5029295496591392,
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  "samples": 66,
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  "loss": 0.4508178895194961,
  "samples": 109,
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  "client_id": "client_44",
  "loss": 0.6560402113888992,
  "samples": 68,
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  "loss": 0.3009254180090019,
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  "samples": 46,
  "epochs": 1
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  "loss": 0.5434186721726524,
  "samples": 50,
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  "client_id": "client_49",
  "loss": 0.343941355340115,
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  "client_id": "client_53",
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  "samples": 84,
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},
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  "client_id": "client_61",
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"samples": 75,
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  "loss": 0.3316252920652513,
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  "samples": 53,
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  "loss": 0.5734168763888122,
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  "loss": 0.484506573418634,
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  "loss": 0.448076118678454,
  "samples": 80,
  "epochs": 1
},
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  "client_id": "client_69",
  "loss": 0.6032384747069421,
  "samples": 42,
  "epochs": 1
},
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  "loss": 0.43450612247381465,
  "samples": 71,
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  "loss": 0.4844331405329872,
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  "client_id": "client_72",
  "loss": 0.5552555824624644,
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  "epochs": 1
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  "samples": 55,
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{
  "client_id": "client_74",
  "loss": 0.5408998109461974,
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},
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  "loss": 0.42813212624594094,
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  "epochs": 1
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  "loss": 0.367722632102926,
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  "client_id": "client_80",
  "loss": 0.5100906479775711,
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  "client_id": "client_81",
  "loss": 0.43214475362939014,
  "samples": 56,
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  "samples": 48,
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  "loss": 0.5334243685131103,
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{
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  "samples": 96,
  "epochs": 1
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{
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  "samples": 41,
  "epochs": 1
},
{
  "client_id": "client_86",
  "loss": 0.5843712745868236,
  "samples": 52,
  "epochs": 1
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{
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  "loss": 0.29684889300147743,
  "samples": 35,
  "epochs": 1
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    "client_id": "client_88",
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    "loss": 0.42387827015921875,
    "samples": 71,
    "epochs": 1
  },
  {
    "client_id": "client_90",
    "loss": 0.288091742329252,
    "samples": 85,
    "epochs": 1
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  {
    "client_id": "client_91",
    "loss": 0.5451559892058415,
    "samples": 67,
    "epochs": 1
  },
  {
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    "loss": 0.37500095988893195,
    "samples": 64,
    "epochs": 1
  },
  {
    "client_id": "client_93",
    "loss": 0.6325860734870078,
    "samples": 55,
    "epochs": 1
  },
  {
    "client_id": "client_94",
    "loss": 0.5500694445182714,
    "samples": 37,
    "epochs": 1
  },
  {
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    "loss": 0.6963010227086347,
    "samples": 46,
    "epochs": 1
  },
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    "loss": 0.4793703150252636,
    "samples": 70,
    "epochs": 1
  },
  {
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    "loss": 0.5304642051544025,
    "samples": 49,
    "epochs": 1
  },
  {
    "client_id": "client_98",
    "loss": 0.5343927648539772,
    "samples": 85,
    "epochs": 1
  },
  {
    "client_id": "client_99",
    "loss": 0.3575260192495336,
    "samples": 76,
    "epochs": 1
  }
],
"timestamp": "2025-12-02T00:56:40.271206"
},
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{
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  "train_loss": 0.46112935484812767,
  "test_metrics": {
    "mse": 0.4507429627388681,
    "mae": 0.45094150242634196,
    "accuracy": 0.5492468134414832,
    "samples": 1726,
    "NDCG@10": 0.0,
    "Hit@10": 0.08,
    "Precision@10": 0.01,
    "Recall@10": 0.00982464846980976,
    "num_users_evaluated": 50
  },
  "aggregation": {
    "status": "aggregated",
    "round": 121,
    "num_clients": 97,
    "total_samples": 6706
  },
  "client_metrics": [
    {
      "client_id": "client_1",
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File: results_old_broken/dp_1_alpha_0.5_dim_16_clients_100_seed_42_summary.csv

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Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients,Total_Samples
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File: results_old_broken/dp_1_alpha_0.5_dim_16_clients_100_seed_456.json

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    "num_users_evaluated": 50
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  "samples": 56,
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      "loss": 0.5128831643908891,
      "samples": 85,
      "epochs": 1
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      "client_id": "client_99",
      "loss": 0.3577028015999562,
      "samples": 76,
      "epochs": 1
    }
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  "timestamp": "2025-12-02T05:56:59.544539"
}
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  "accuracy": 0.5492468134414832,
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  "Hit@10": 0.08,
  "Precision@10": 0.008,
  "Recall@10": 0.007964697138610182,
  "num_users_evaluated": 50
},
"client_metrics": [],
"mobile_metrics": []
}

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File: results_old_broken/dp_1_alpha_0.5_dim_16_clients_100_seed_456_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients>Total_Samples
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44,0.5492468134414832,97,6706
4,0.4741529697509919,0.0,0.08,0.008,0.007570431905635271,0.45075429093658165,0.4509740732221725,0.549246813
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5,0.4723836951081728,0.0,0.06,0.006000000000000001,0.007409141583054626,0.4507526739447584,0.45097158129218
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6,0.46314159843503644,0.0,0.1,0.01,0.0076319703671737325,0.4507551985433452,0.45098258556662124,0.549246813
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7,0.46088863605206043,0.0,0.12,0.012000000000000002,0.0078188862550242,0.4507609536722389,0.450990358620087
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8,0.4627939696796279,0.0,0.1,0.01,0.008151613026460648,0.45075994840502603,0.4509947899457063,0.54924681344
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4832,97,6706
10,0.47212528499010786,0.0,0.08,0.008,0.007964697138610182,0.45075876971108886,0.4510075148121645,0.5492468
134414832,97,6706
FINAL,,0.0,0.08,0.008,0.007964697138610182,0.45075876971108886,0.4510075148121645,0.5492468134414832,,

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File: results_old_broken/dp_2_alpha_0.5_dim_16_clients_100_seed_123.json

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    "dp_epsilon": 2,
    "use_dp": true,
    "dp_sigma": 1.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 1,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 123
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  "rounds": [
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        "mse": 0.45074248576357634,
        "mae": 0.4509111506085269,
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        "Hit@10": 0.04,
        "Precision@10": 0.004,
        "Recall@10": 0.0026050420168067224,
        "num_users_evaluated": 50
      },
      "aggregation": {
        "status": "aggregated",
        "round": 100,
        "num_clients": 97,
        "total_samples": 6706
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          "loss": 0.5467943050358842,
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        {
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          "loss": 0.6460271823842054,
          "samples": 53,
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        {
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          "loss": 0.6086568591680607,
          "samples": 92,
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        {
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          "loss": 0.45314098589288593,
          "samples": 56,
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          "client_id": "client_5",
          "loss": 0.4891374964244081,
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}
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        "loss": 0.409483872453384,
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    "accuracy": 0.5492468134414832,
    "samples": 1726,
    "NDCG@10": 0.0,
    "Hit@10": 0.08,
    "Precision@10": 0.008,
    "Recall@10": 0.0028012459464072364,
    "num_users_evaluated": 50
},
"client_metrics": [],
"mobile_metrics": []
}

```

File: results_old_broken/dp_2_alpha_0.5_dim_16_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients,Total_Samples
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2,0.47920174649686825,0.0,0.02,0.002,0.0014285714285714286,0.45074996825303293,0.4509178970531406,0.5492468
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4,0.4713242784026639,0.0,0.06,0.0060000000000000001,0.0021560846560846557,0.45074582196608065,0.450918640764
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File: results_old_broken/dp_2_alpha_0.5_dim_16_clients_100_seed_42.json

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        "num_clients": 100,
        "alpha": 0.5,
        "dp_epsilon": 2,
        "use_dp": true,
        "dp_sigma": 1.0,
        "dp_clip_norm": 1.0,
        "num_rounds": 10,
        "local_epochs": 1,
        "learning_rate": 0.01,
        "batch_size": 32,
    }
}

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      "Precision@10": 0.002,
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      "num_users_evaluated": 50
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  "samples": 52,
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{
  "client_id": "client_75",
  "loss": 0.392328037587222,
  "samples": 113,
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"epochs": 1
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  "samples": 92,
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  "loss": 0.5342578898859018,
  "samples": 47,
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  "samples": 69,
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  "loss": 0.3542034041375514,
  "samples": 96,
  "epochs": 1
},
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  "loss": 0.583276457018998,
  "samples": 41,
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  "samples": 52,
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},
{
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  "samples": 35,
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},
{
  "client_id": "client_88",
  "loss": 0.4866970579200486,
  "samples": 51,
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},
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    "loss": 0.4610151403238259,
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  {
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    "loss": 0.277223017791302,
    "samples": 85,
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  {
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    "loss": 0.3437296842610933,
    "samples": 67,
    "epochs": 1
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  {
    "client_id": "client_92",
    "loss": 0.3751365901903032,
    "samples": 64,
    "epochs": 1
  },
  {
    "client_id": "client_93",
    "loss": 0.6327166970007213,
    "samples": 55,
    "epochs": 1
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  {
    "client_id": "client_94",
    "loss": 0.3811724738772831,
    "samples": 37,
    "epochs": 1
  },
  {
    "client_id": "client_95",
    "loss": 0.6762022869350961,
    "samples": 46,
    "epochs": 1
  },
  {
    "client_id": "client_96",
    "loss": 0.47897151041411384,
    "samples": 70,
    "epochs": 1
  },
  {
    "client_id": "client_97",
    "loss": 0.5302612216381116,
    "samples": 49,
    "epochs": 1
  },
  {
    "client_id": "client_98",
    "loss": 0.529407930731948,
    "samples": 85,
    "epochs": 1
  },
  {
    "client_id": "client_99",
    "loss": 0.34033200721296475,
    "samples": 76,
    "epochs": 1
  }
],
"timestamp": "2025-12-01T21:19:14.173383"
}
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"final_metrics": {
  "mse": 0.4507887872456123,
  "mae": 0.451024299701147,
  "accuracy": 0.5492468134414832,
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"samples": 1726,
"NDCG@10": 0.0,
"Hit@10": 0.02,
"Precision@10": 0.002,
"Recall@10": 6.153846153846154e-05,
"num_users_evaluated": 50
},
"client_metrics": [],
"mobile_metrics": []
}
```

File: results_old_broken/dp_2_alpha_0.5_dim_16_clients_100_seed_42_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients,Total_Samples
1,0.47154936668868247,0.0,0.02,0.002,0.0007142857142857143,0.4507739180623449,0.45095272439932355,0.5492468
134414832,97,6706
2,0.47049358576060213,0.0,0.06,0.006000000000000001,0.0012557161148710444,0.4507738812497427,0.450959739375
6393,0.5492468134414832,97,6706
3,0.47401128761814265,0.0,0.12,0.012000000000000002,0.0033368074373416774,0.4507859865870426,0.450970635353
34475,0.5492468134414832,97,6706
4,0.46554954825525274,0.0,0.1,0.012,0.0026621103034896137,0.4507883632445529,0.4509803234909252,0.549246813
4414832,97,6706
5,0.4684146945661216,0.0,0.06,0.008,0.0008887112887112889,0.4507865871228321,0.4509841978619024,0.549246813
4414832,97,6706
6,0.4679101545192908,0.0,0.04,0.004,0.0005674325674325675,0.45078819444530394,0.45099301299518396,0.5492468
134414832,97,6706
7,0.4667230535444216,0.0,0.02,0.002,0.0003076923076923077,0.4507800112041372,0.45099213336047267,0.54924681
34414832,97,6706
8,0.4734137805305281,0.0,0.02,0.002,0.00016129032258064516,0.45077989309991634,0.45100471014772064,0.549246
8134414832,97,6706
9,0.47241545603553714,0.0,0.02,0.002,0.0014285714285714286,0.4507790493108721,0.45101515525738306,0.5492468
134414832,97,6706
10,0.4669357790063967,0.0,0.02,0.002,6.153846153846154e-05,0.4507887872456123,0.451024299701147,0.549246813
4414832,97,6706
FINAL,,0.0,0.02,0.002,6.153846153846154e-05,0.4507887872456123,0.451024299701147,0.5492468134414832,,
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File: results_old_broken/dp_2_alpha_0.5_dim_16_clients_100_seed_456.json

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{
  "experiment_id": "dp_2_alpha_0.5_dim_16_clients_100_seed_456",
  "timestamp": "2025-12-01T23:03:00.940371",
  "config": {
    "num_users": 50,
    "num_items": 4032,
    "embedding_dim": 16,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 2,
    "use_dp": true,
    "dp_sigma": 1.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 1,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 0.4694214432423877,
      "test_metrics": {
        "mse": 0.45075887047954394,
        "mae": 0.4509323502470969,
        "accuracy": 0.5492468134414832,
        "samples": 1726,
        "NDCG@10": 0.0,
        "Hit@10": 0.08,
        "Precision@10": 0.01,
        "Recall@10": 0.0032724502481749095,
        "num_users_evaluated": 50
      },
      "aggregation": {
        "status": "aggregated",

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"round": 110,  
"num_clients": 97,  
"total_samples": 6706  
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"client_metrics": [  
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    "loss": 0.5513147339365966,  
    "samples": 60,  
    "epochs": 1  
  },  
  {  
    "client_id": "client_2",  
    "loss": 0.6703042612719721,  
    "samples": 53,  
    "epochs": 1  
  },  
  {  
    "client_id": "client_3",  
    "loss": 0.6084968803548098,  
    "samples": 92,  
    "epochs": 1  
  },  
  {  
    "client_id": "client_4",  
    "loss": 0.45319330384367856,  
    "samples": 56,  
    "epochs": 1  
  },  
  {  
    "client_id": "client_5",  
    "loss": 0.47048835171292813,  
    "samples": 81,  
    "epochs": 1  
  },  
  {  
    "client_id": "client_6",  
    "loss": 0.45305838879768695,  
    "samples": 64,  
    "epochs": 1  
  },  
  {  
    "client_id": "client_7",  
    "loss": 0.3406983571058526,  
    "samples": 75,  
    "epochs": 1  
  },  
  {  
    "client_id": "client_8",  
    "loss": 0.3858103607174194,  
    "samples": 122,  
    "epochs": 1  
  },  
  {  
    "client_id": "client_9",  
    "loss": 0.5549619747591477,  
    "samples": 98,  
    "epochs": 1  
  },  
  {  
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    "loss": 0.41669252747380925,  
    "samples": 66,  
    "epochs": 1  
  },  
  {  
    "client_id": "client_11",  
    "loss": 0.4998866353420598,  
    "samples": 40,  
    "epochs": 1  
  },  
  {  
    "client_id": "client_12",  
    "loss": 0.3838460528870204,
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"samples": 105,
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{
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  "loss": 0.646586214068038,
  "samples": 42,
  "epochs": 1
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  "loss": 0.39193368801363027,
  "samples": 86,
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{
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  "loss": 0.7182122061729169,
  "samples": 68,
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  "client_id": "client_16",
  "loss": 0.36228536767494474,
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  "loss": 0.42181922266029676,
  "samples": 34,
  "epochs": 1
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{
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  "loss": 0.38991934971489206,
  "samples": 82,
  "epochs": 1
},
{
  "client_id": "client_20",
  "loss": 0.49524555983460483,
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  "loss": 0.44191605023302577,
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  "samples": 68,
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  "epochs": 1
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  "loss": 0.44846540641782695,
  "samples": 66,
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  "loss": 0.48423480617666426,
  "samples": 36,
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{
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  "samples": 68,
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  "loss": 0.45004190354197543,
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  "samples": 80,
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},
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  "loss": 0.39690263095733014,
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},
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  "client_id": "client_70",
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  "client_id": "client_73",
  "loss": 0.4238939251971261,
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  "epochs": 1
},
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  "client_id": "client_74",
  "loss": 0.5311035838592187,
  "samples": 52,
  "epochs": 1
},
{
  }
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"client_id": "client_75",
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  "samples": 92,
  "epochs": 1
},
{
  "client_id": "client_77",
  "loss": 0.5166020790593425,
  "samples": 47,
  "epochs": 1
},
{
  "client_id": "client_79",
  "loss": 0.45874008146163947,
  "samples": 70,
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  "loss": 0.5557217036989717,
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{
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{
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{
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{
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  "samples": 41,
  "epochs": 1
},
{
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  "loss": 0.5466320463727737,
  "samples": 52,
  "epochs": 1
},
{
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  "loss": 0.44776272737348094,
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  },
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    "samples": 85,
    "epochs": 1
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  {
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    "epochs": 1
  },
  {
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    "loss": 0.37503016615026874,
    "samples": 64,
    "epochs": 1
  },
  {
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    "loss": 0.6262750474467352,
    "samples": 55,
    "epochs": 1
  },
  {
    "client_id": "client_94",
    "loss": 0.7187895757172116,
    "samples": 37,
    "epochs": 1
  },
  {
    "client_id": "client_95",
    "loss": 0.7165684991471621,
    "samples": 46,
    "epochs": 1
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  {
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    "loss": 0.4791067533080282,
    "samples": 70,
    "epochs": 1
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    "loss": 0.5439303181875813,
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    "epochs": 1
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  {
    "client_id": "client_98",
    "loss": 0.5020484542255529,
    "samples": 85,
    "epochs": 1
  },
  {
    "client_id": "client_99",
    "loss": 0.40985851732323814,
    "samples": 76,
    "epochs": 1
  }
],
"timestamp": "2025-12-01T23:13:19.650789"
},
{
  "round": 2,
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File: results_old_broken/dp_2_alpha_0.5_dim_16_clients_100_seed_456_summary.csv

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Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients,Total_Samples
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File: results_old_broken/dp_4_alpha_0.5_dim_16_clients_100_seed_123.json

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    "samples": 1726,
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    "Hit@10": 0.08,
    "Precision@10": 0.008,
    "Recall@10": 0.0068780823025305195,
    "num_users_evaluated": 50
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"client_metrics": [],
"mobile_metrics": []
}
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File: results_old_broken/dp_4_alpha_0.5_dim_16_clients_100_seed_123_summary.csv

Round	Train_Loss	NDCG@10	Hit@10	Precision@10	Recall@10	MSE	MAE	Accuracy	Num_Clients	Total_Samples
1	0.4712409314616096	0.0	0.06	0.006000000000000001	0.005350877192982456	0.45076236995595353	0.4511182404890	536	0.5492468134414832	97, 6706
2	0.4769995293972792	0.0	0.06	0.006000000000000001	0.0030572862582208377	0.45074279290361724	0.451348422990	88956	0.5492468134414832	97, 6706
3	0.46498163522706243	0.0	0.04	0.004	0.001394871794871795	0.45067438873465615	0.4515471154541157	0.5492468134414832	97, 6706	
4	0.47168759237399266	0.0	0.12	0.012000000000000002	0.0029677988677988677	0.4506462650116736	0.451787504370	76787	0.5492468134414832	97, 6706
5	0.4741665503954508	0.0	0.08	0.008	0.0009778145391048617	0.45068097059309553	0.45201196261656795	0.5492468134414832	97, 6706	
6	0.46540830261194255	0.0	0.02	0.002	0.00016129032258064516	0.45054934523473056	0.45215149854038983	0.5492468134414832	97, 6706	
7	0.46743645094429576	0.0	0.06	0.006000000000000001	0.0038649940262843485	0.4505484994427492	0.452372406200	35146	0.5492468134414832	97, 6706
8	0.4769480371157906	0.0	0.02	0.002	0.00016129032258064516	0.4506816021398763	0.4525374426902515	0.5492468134414832	97, 6706	
9	0.4738905233140998	0.0	0.06	0.006000000000000001	0.0036815395437644455	0.4507269275008872	0.4527709116664	988	0.5492468134414832	97, 6706
10	0.4678088589476375	0.0	0.08	0.008	0.0068780823025305195	0.45060178012450075	0.45296686908461126	0.5492468134414832	97, 6706	
FINAL		0.0	0.08	0.008	0.0068780823025305195	0.45060178012450075	0.45296686908461126	0.5492468134414832		